Lesson notes · for you

A device that remembers

IoT · Level 3 · Lesson 3

Materials
A thermometer, or a stick and a ruler for measuring shadows; Paper and a pen for a table; A clock or a phone; Optional: the free phyphox app, which logs a phone's own sensors

Learning goals

- Decide how often a device should take readings for a job.
- Explain why every reading is a little wrong, and how averaging helps and costs.
- Plan a data log with times and units.

Activity: Be a data logger

1. Choose one thing: the temperature in the shade (with a thermometer), or the length of a stick's shadow (with a ruler). Pick one spot and use it all day.
2. Draw a table with six columns: date, time, reading, unit, place and notes. Fill in the date and the place first.
3. Every hour from 7:00 to 17:00, take the reading at the same spot and write it straight into the table, with its unit. Missed an hour? Leave the row empty and say why.
4. Draw a line chart: the hours along the bottom, the readings up the side. Join the points.
5. Look for anything that can't be right: a sudden jump, a missing unit, a shadow longer at noon than in the morning. Circle it, and write what you'd do: check, measure again, or remove it with a note.

Keep the thermometer in the shade, never in direct sun. Never look straight at the sun when you measure shadows, and don't leave a phone in hot sun.

Discussion

- What could make your readings a little wrong?
- When would one reading a day be enough, and when would it be far too few?
- Why keep the raw data after you've cleaned it?

Check yourself

- Name a job that needs a reading every few seconds, and one that needs one a day.

Try it this week
Be a data logger: For one day, take a reading every hour: a thermometer in the shade, or the length of a stick's shadow in the sun. Write the date, time, reading, unit and place in a table. Then draw a chart, mark any reading that can't be right, and write what you would do about it.

Worksheet

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Name _____ Class _____ Date _____

1 **A gas alarm averages its last 10 minutes of readings, to hide noise. What is the problem?**

- ◯ It gives more false alarms when people cook
- ◯ Ten minutes of readings fill up its memory
- ◯ It reacts minutes too late to a real leak

2 **Match each job to a sensible reading interval.**

A rain gauge for farmers	• • • •	Every 10 to 30 seconds
Soil moisture in the school garden	• • • •	Every few seconds
A bus's position on a live map	• • • •	Once a day
A smoke alarm	• • • •	Every hour

3 **Three classroom readings: 29.5, 30.5 and 30 °C.**

Their average is _____ °C. If a glitch of 85 replaces the 30, the average jumps to about _____ °C.

Word bank: 85 · 38 · 30 · 48

4 **Reading a thermometer every second instead of every minute makes each reading more accurate.**

- ◯ True
- ◯ False

5 **You're planning a log of the school water tank. What must every row have?**

- ☐ The sensor's price
- ☐ The weather forecast
- ☐ The date and time
- ☐ The reading with its unit (% or litres)
- ☐ Which tank, if there's more than one

✎ **What would you like to log for a week at home or at school? How often would you read it, and what could make a reading wrong?**

Check your answers

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1 **It reacts minutes too late to a real leak**

A ten-minute average still remembers clean air, so a real leak takes minutes to push it over the threshold.

2 **A rain gauge for farmers — Once a day**
Soil moisture in the school garden — Every hour
A bus's position on a live map — Every 10 to 30 seconds
A smoke alarm — Every few seconds

Match the interval to how fast things change and how quickly you must act.

3 **Their average is 30 °C. If a glitch of 85 replaces the 30, the average jumps to about 48 °C.**

(29.5 + 30.5 + 30) ÷ 3 = 30. With the glitch: (29.5 + 30.5 + 85) ÷ 3 ≈ 48, far from the real room.

4 **False**

Each reading has the same noise however often you take it.

5 **The date and time; The reading with its unit (% or litres); Which tank, if there's more than one**

Time, unit and place turn a number into a fact someone can use: "40% at 7:00 in the kitchen tank".